

Functions of Two Variables — Question Bank

Limits, Continuity and Partial Derivatives

Engineering Mathematics–I (CSE-1001)

BTech 1st Year, IIIT Bhopal

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Instructions

This bank accompanies the note *Functions of Two Variables*, the second note of Module 2. Read that note, and the companion note *The Mean Value Theorems*, before attempting it.

How the parts are meant to be used:

1. **Part A (MCQ)** — rapid revision. One minute per question.
2. **Part B (True or False)** — every answer must carry a reason or a counterexample.
3. **Part C (Short answer)** — routine computations: one limit, one continuity check or one derivative per question.
4. **Part D (Long answer)** — definitions, proofs and the longer multi-case problems.
5. **Part E (Critical thinking)** — traps, counterexamples and faulty arguments to be diagnosed.

The habit this topic demands. Before writing anything about a limit in two variables, decide which of two jobs you are doing.

- *Disproving* a limit: find two paths giving different values. One counterexample finishes the question.
- *Proving* a limit: you must bound $|f - L|$ by a quantity depending on r **alone**. No number of paths will ever do this.

Mixing these up is the single largest source of lost marks in this topic.

Answers and full worked solutions are issued as a separate document, *Functions of Two Variables — Answer Key*.

Part A: Multiple Choice

30 questions

- Q1.** In the definition of $\lim_{(x,y) \rightarrow (a,b)} f(x,y) = L$, the condition on (x,y) is that it lies in a small punctured
- (a) interval (b) disc (c) square (d) ray
- Q2.** A limit in two variables exists only if the same value is obtained along
- (a) the two axes (b) all straight lines (c) all parabolas (d) every possible path
- Q3.** $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2}$
- (a) does not exist
(b) equals 0
(c) equals 1
(d) equals -1
- Q4.** For $f = \frac{xy}{x^2 + y^2}$, the value along the path $y = x$ (with $x \neq 0$) is
- (a) 0 (b) 1 (c) $\frac{1}{2}$ (d) 2
- Q5.** $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2}$
- (a) is 0 (b) is $\frac{1}{2}$ (c) is 1 (d) does not exist
- Q6.** For $f = \frac{x^2 y}{x^4 + y^2}$, the limit along every straight line through the origin is
- (a) $\frac{1}{2}$ (b) 0 (c) 1 (d) different for each line
- Q7.** For the same $f = \frac{x^2 y}{x^4 + y^2}$, approaching along $y = x^2$ gives
- (a) 0 (b) 1 (c) $\frac{1}{2}$ (d) ∞
- Q8.** Agreement of the limit along every straight line through a point proves
- (a) nothing about the existence of the limit
(b) that the limit exists
(c) that the function is continuous
(d) that the partial derivatives exist
- Q9.** Substituting $x = r \cos \theta$, $y = r \sin \theta$ converts $(x,y) \rightarrow (0,0)$ into
- (a) $\theta \rightarrow 0$ (b) $r \rightarrow \infty$ (c) $\theta \rightarrow 2\pi$ (d) $r \rightarrow 0^+$
- Q10.** A polar computation establishes a limit only if the final bound depends on
- (a) θ alone (b) r alone (c) both r and θ (d) neither

- Q11.** In polar form $\frac{x^2y}{x^2+y^2} = r \cos^2 \theta \sin \theta$, so the limit at the origin is
 (a) 1 (b) does not exist (c) 0 (d) $\frac{1}{2}$
- Q12.** The two repeated limits of f at (a, b) are
 (a) $\lim_{x \rightarrow a} \lim_{y \rightarrow b} f$ and $\lim_{y \rightarrow b} \lim_{x \rightarrow a} f$
 (b) $\lim_{(x,y) \rightarrow (a,b)} f$ taken twice
 (c) the limits along the two axes
 (d) the partial derivatives
- Q13.** If the two repeated limits exist and are different, then the double limit
 (a) equals their average (b) does not exist (c) equals the larger (d) equals 0
- Q14.** If the two repeated limits exist and are equal, the double limit
 (a) must exist (b) must equal them (c) must be 0 (d) may still fail to exist
- Q15.** For $f = \frac{x-y}{x+y}$, the repeated limits at the origin are
 (a) both 0 (b) both 1 (c) 1 and -1 (d) both undefined
- Q16.** f is continuous at (a, b) if $f(a, b)$ is defined, the limit exists, and
 (a) the two are equal (b) the limit is 0 (c) f is bounded (d) f_x and f_y exist
- Q17.** The function $f = \frac{x^2y}{x^2+y^2}$ with $f(0, 0) = 0$ is
 (a) discontinuous at the origin (b) undefined at the origin (c) unbounded near the origin (d) continuous at the origin
- Q18.** The function $f = \frac{x^2-y^2}{x^2+y^2}$ with $f(0, 0) = 0$ has at the origin a discontinuity that is
 (a) removable (b) not removable (c) of the jump type (d) at infinity
- Q19.** Compared with one variable, checking continuity in two variables is harder because
 (a) there are infinitely many directions of approach
 (b) the function may be unbounded
 (c) limits need not be unique
 (d) derivatives are harder to compute
- Q20.** $\frac{\partial f}{\partial x}$ is computed by differentiating with respect to x while treating y as
 (a) zero (b) a function of x (c) a constant (d) infinite
- Q21.** Geometrically, $f_x(a, b)$ is the slope of the curve cut from the surface by the plane
 (a) $x = a$ (b) $z = 0$ (c) $x = y$ (d) $y = b$

- Q22.** For $f = x^2y + y^3$, f_y equals
 (a) $2xy$ (b) $x^2 + 3y^2$ (c) $x^2 + y^2$ (d) $3y^2$
- Q23.** For $f = e^{xy}$, f_x equals
 (a) e^{xy} (b) xe^{xy} (c) ye^{xy} (d) xye^{xy}
- Q24.** For $f = \tan^{-1}\left(\frac{y}{x}\right)$, f_x equals
 (a) $\frac{-y}{x^2 + y^2}$ (b) $\frac{y}{x^2 + y^2}$ (c) $\frac{x}{x^2 + y^2}$ (d) $\frac{-x}{x^2 + y^2}$
- Q25.** For $f = \frac{xy}{x^2 + y^2}$ with $f(0, 0) = 0$, the partial derivatives at the origin
 (a) do not exist
 (b) both exist and equal 0
 (c) both exist and equal $\frac{1}{2}$
 (d) exist only in the x direction
- Q26.** The existence of both partial derivatives at a point implies
 (a) continuity there (b) differentiability there (c) boundedness there (d) neither continuity nor differentiability
- Q27.** In the notation f_{xy} used in your notes, the differentiation is performed
 (a) with respect to x first, then y
 (b) twice with respect to x
 (c) with respect to y first, then x
 (d) in either order, as they always agree
- Q28.** Clairaut's Theorem guarantees $f_{xy} = f_{yx}$ provided
 (a) f is a polynomial
 (b) both partial derivatives exist
 (c) the mixed partials are continuous near the point
 (d) f is bounded
- Q29.** A function f is homogeneous of degree n if, for all $t > 0$,
 (a) $f(tx, ty) = nf(x, y)$ (b) $f(tx, ty) = t^n f(x, y)$ (c) $f(x + t, y + t) = t^n f(x, y)$ (d) $f(tx, ty) = t f(x, y)^n$
- Q30.** Euler's Theorem states that for such a function
 (a) $f_x + f_y = nf$ (b) $xf_x - yf_y = nf$ (c) $xf_x + yf_y = nf$ (d) $x^2f_{xx} + y^2f_{yy} = nf$

Part B: True or False

10 questions — give a reason or a counterexample

- Q1.** If $f(x, 0) \rightarrow 0$ and $f(0, y) \rightarrow 0$ as the variable tends to 0, then $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0$.
- Q2.** If the limit along every straight line through $(0, 0)$ is L , then the double limit is L .
- Q3.** If $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ exists, then f is bounded near the origin.
- Q4.** A polar computation yielding the bound $|f| \leq r/|\sin \theta|$ establishes that the limit is 0.
- Q5.** If both repeated limits at $(0, 0)$ exist and are equal, the double limit exists.
- Q6.** If the double limit exists, then both repeated limits exist.
- Q7.** If f is continuous at (a, b) , then $f_x(a, b)$ and $f_y(a, b)$ exist.
- Q8.** If $f_x(a, b)$ and $f_y(a, b)$ exist, then f is continuous at (a, b) .
- Q9.** For every function of two variables, $f_{xy} = f_{yx}$.
- Q10.** If f is homogeneous of degree n , then so is f_x , of degree $n - 1$.

Part C: Short Answer

18 questions

- Q1.** State the ε - δ definition of $\lim_{(x,y) \rightarrow (a,b)} f(x, y) = L$, and say in one sentence how it differs from the one-variable definition.
- Q2.** Show that $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2}$ does not exist.
- Q3.** Show that $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2}$ does not exist.
- Q4.** Show that $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^4 + y^2}$ does not exist, and explain why the straight-line paths were not enough.
- Q5.** Show that $\lim_{(x,y) \rightarrow (0,0)} \frac{x^3 y}{x^6 + y^2}$ does not exist. *Hint: the straight lines all give 0; look at the exponents in the denominator to choose a better path.*
- Q6.** Decide whether $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2}{x^2 + y^2}$ exists, and evaluate it if it does.
- Q7.** Evaluate $\lim_{(x,y) \rightarrow (0,0)} \frac{x^3}{x^2 + y^2}$ using polar coordinates.
- Q8.** Evaluate $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{\sqrt{x^2 + y^2}}$.
- Q9.** Evaluate $\lim_{(x,y) \rightarrow (0,0)} (x^2 + y^2) \ln(x^2 + y^2)$.

- Q10.** Compute both repeated limits of $f = \frac{x-y}{x+y}$ at the origin, and state what they tell you about the double limit.
- Q11.** Compute both repeated limits of $f = \frac{x^2y}{x^4+y^2}$ at the origin. Do they settle the question of the double limit?
- Q12.** Is $f = \frac{x^2y}{x^2+y^2}$, with $f(0,0) = 0$, continuous at the origin? Justify.
- Q13.** Is $f = \frac{x^2-y^2}{x^2+y^2}$, with $f(0,0) = 0$, continuous at the origin? Can the discontinuity be removed by redefining $f(0,0)$?
- Q14.** Find f_x and f_y from first principles for $f(x,y) = x^2y + y^3$.
- Q15.** Find all first-order partial derivatives of $f(x,y,z) = x^2y + yz^3 + xz$.
- Q16.** Find all first-order partial derivatives of each of
- (i) $f(x,y,z) = e^{xyz}$ (ii) $f(x,y,z) = \ln(x^2 + y^2 + z^2)$.
- Q17.** For $f(x,y,z) = \ln(xy + z^2)$, compute f_y and then f_{yx} , showing the quotient rule in full.
- Q18.** Verify Euler's Theorem for $f(x,y) = \frac{x^3 + y^3}{x + y}$, stating the degree first.

Part D: Long Answer

10 questions

- Q1.** Explain carefully why a limit in two variables is harder to establish than a limit in one variable. Your answer should contrast the two cases, state the path principle, and say explicitly what the path principle can and cannot prove.
- Q2.** For $f(x,y) = \frac{x^2y}{x^4+y^2}$:
- (i) show that the limit along every line $y = mx$ is 0;
 - (ii) show that the limit along $y = x^2$ is $\frac{1}{2}$;
 - (iii) conclude, and explain how the exponents in the denominator tell you which curve to try.
- Q3.** State the polar-coordinate method for proving that a two-variable limit exists, and explain precisely why the final bound must be free of θ . Illustrate with one limit that the method settles and one that it does not.
- Q4.** Define the two repeated limits and the double limit. State correctly the relationship between them, prove the part you can, and give counterexamples showing that the two converse statements both fail.

Q5. Define continuity for a function of two variables. Determine, with full justification, whether each of the following is continuous at the origin (each defined to be 0 there):

$$(i) \frac{x^2 y}{x^2 + y^2} \quad (ii) \frac{x^2 - y^2}{x^2 + y^2} \quad (iii) \frac{xy}{x^2 + y^2}$$

Q6. Define the partial derivatives of $f(x, y)$ as limits. Explain their geometric meaning in terms of slices of the surface $z = f(x, y)$, and compute both from first principles for $f(x, y) = xy^2$.

Q7. Let

$$f(x, y) = \begin{cases} \frac{xy}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

Show that $f_x(0, 0)$ and $f_y(0, 0)$ both exist, that f is not continuous at the origin, and explain why this does not contradict anything you know from one variable.

Q8. Let

$$f(x, y) = \begin{cases} \frac{xy(x^2 - y^2)}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

Compute $f_x(0, y)$ and $f_y(x, 0)$, hence show that $f_{xy}(0, 0)$ and $f_{yx}(0, 0)$ differ. State which is -1 and which is $+1$ under the convention of your notes, and say why the convention must be stated.

Q9. State Clairaut's Theorem. Explain why the function of the previous question is not a counterexample to it, identifying exactly which hypothesis fails.

Q10. Define a homogeneous function of degree n . State and prove Euler's Theorem, and deduce the second-order form

$$x^2 f_{xx} + 2xy f_{xy} + y^2 f_{yy} = n(n-1)f.$$

Verify the first-order result for $f = \tan^{-1}(y/x)$.

Part E: Critical Thinking

8 questions

Q1. A student writes:

“Along $y = 0$ the limit is 0; along $x = 0$ it is 0; along $y = mx$ it is 0 for every m . Since all paths give 0, the limit is 0.”

Identify the flaw in the final sentence, and produce a function for which every line gives 0 but the limit does not exist.

Q2. A set of lecture notes contains both of the following statements:

“If all paths give the same result, the limit exists.”

and, four pages later, the worked example showing that $\frac{x^2y}{x^4+y^2}$ has equal limits along all lines but no limit. Are the two statements compatible? Rewrite the first so that it is both true and usable.

Q3. A student evaluates $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2+y^2}$ in polar coordinates, obtains $\frac{1}{2} \sin 2\theta$, and concludes that the limit is $\frac{1}{2}$ “because $\sin 2\theta$ averages to a constant”. Diagnose the error, and state what the polar form actually proves here.

Q4. A set of lecture notes states:

“Theorem. If the double limit $\lim_{(x,y) \rightarrow (a,b)} f(x,y)$ exists, then both repeated limits exist and are equal to it.”

- (i) Show that this is false, using $f(x,y) = x \sin(1/y)$ for $y \neq 0$ and $f(x,0) = 0$.
- (ii) Repair the statement by adding the missing hypothesis.
- (iii) Explain why the *converse* also fails, with a different example.

Q5. A student argues: “ f is discontinuous at the origin, so its partial derivatives cannot exist there.” Decide whether this is right. If it is wrong, give an explicit counterexample and explain which one-variable fact is being misremembered.

Q6. A student computes, for $f(x,y,z) = \ln(xy+z^2)$ with $f_y = \frac{x}{xy+z^2}$,

$$f_{yx} = \frac{(xy+z^2) \cdot 0 - x \cdot y}{(xy+z^2)^2} = \frac{-xy}{(xy+z^2)^2}.$$

- (i) Identify the error and give the correct answer.
- (ii) The wrong answer is symmetric in x and y , as Clairaut’s Theorem demands. Explain why the symmetry check therefore fails to detect this error, and say what that tells you about using symmetry as a check.

Q7. Two textbooks both compute the mixed partials of $\frac{xy(x^2-y^2)}{x^2+y^2}$ at the origin. One reports $f_{xy}(0,0) = +1$, the other $f_{xy}(0,0) = -1$. Both are considered correct. Explain how this is possible, and state what each book must have said earlier in order for its answer to be right.

Q8. A student applies Euler’s Theorem to $f(x,y) = x^2 + y^2 + x$, asserting that it is homogeneous of degree 2, and concludes $xf_x + yf_y = 2f$.

- (i) Verify by direct computation that the conclusion is false.
- (ii) Show from the definition that f is not homogeneous.
- (iii) State what goes wrong if a function is a *sum* of homogeneous functions of different degrees.